MILITARY SPECIFICATION

CANISTER, CHEMICAL PROJECTILE: 8 INCH, QL, XM27

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of filled canister.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-N-411 - Nitrogen, Technical.
BB-H-1168 - Helium, Technical.

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification For.

FSC 1320

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 15-12-162 - Canister, Chemical Projectile: 8 Inch, QL, XM27.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 15-12-162 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components shall conform to the specifications and drawings listed on TDPL 15-12-162 and subsidiary drawings.

3.2 Filling and closing. The canister shall be filled and closed as follows (see 6.3):

(a) Empty canister. Inspect vinyl tape, covering the canister fill port. If tape is punctured or loose so as to prevent an airtight seal, canister shall be removed from filling line for reworking to assure clean, dry canister interior. If tape is acceptable, remove tape and burst disk protective cover and send canister to purge station.

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(b) Purge. Purge the interior of the canister with dry nitrogen, type 1, class 2, grade B of BB-N-411 for 15 seconds at a minimum flow rate of 2 cubic feet per minute (cfm). Immediately after purge cover fill port with clean dry tapered teflon plug to maintain purged atmosphere. NOTE: Cover is unnecessary if purge and fill stations are combined or if movement between them is enclosed with the same dry nitrogen atmosphere.

(c) QL filling. Add .2 percent by weight (.03 pounds ± 10 percent (.13 kilograms)) of dry thickner (Dwg B15-12-208) to the empty canister. Then fill canister with 14.80 ± .15 pounds (6.3 kilograms) of QL (see 6.7). After filling, cover fill port with clean, dry, tapered teflon plug and send canister to next station for closing. NOTE: Plug is unnecessary if movement between stations is enclosed with same dry nitrogen atmosphere.

(d) Simulant TEP/DBA/EA filling. Fill canister with 14.80 ± .15 pounds (6.3 kilograms) of simulant (Dwg 15-12-212). After filling, cover fill port with clean, dry, tapered plug (3.2(c)) and send canister to next station for closing. See note 3.2(c).

(e) Closing (weld). The void over the liquid shall be purged with dry nitrogen, type 1, class 2, grade B of BB-N-411 for 10 seconds at a minimum flow rate of 30 cubic feet per hour (cfh). The weld plug shall be welded in place within 1 minute after the nitrogen purge.

3.3 Leakage. The welded closure shall not leak helium at a rate greater than 1.0×10^{-6} atmospheric cubic centimeters per second under a vacuum of 15 microns of mercury when tested as specified in 4.3.4.1. After testing, burst disk protective cover shall be replaced. The preproduction shall be purged in accordance with 3.2(e) except that helium, grade A of BB-H-1168 shall be used instead of nitrogen.

3.4 Preproduction. Prior to the start of regular production, a preproduction lot of canisters shall be produced in accordance with this specification for examination and test (see 4.3).

3.5 Workmanship. The canisters shall be free from contamination (QL spillage, (see 6.5), foreign matter, rust, oil or grease) and damage such as dents, cracks or deformation which could affect its intended use, and shall conform to the standards of workmanship for parts and assemblies listed in MIL-A-2550.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified

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in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 100 canisters shall be filled and closed using the same methods, materials, equipment and processes as will be used during regular production (see 6.2).

4.3.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.3 Inspection procedures. The sample canisters shall be examined in accordance with the classification of defects (4.4.3.3) and MIL-STD-105. Testing shall be conducted in accordance with 4.3.4 and 4.4.4. The acceptance number is zero.

4.3.4 Test. The test shall be conducted as follows:

4.3.4.1 Leakage. The assembled canister shall be placed under a test head connected to a vacuum system and a mass spectrometer type helium leak detector. The test head shall be lowered over the fill port head. The leak detector shall be calibrated to respond to a leak rate of 1.0×10^{-6} atmospheric cubic centimeters per second of helium when a standard leak is introduced.

4.3.5 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.3 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

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4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. A lot shall consist of canisters of one interfix number, produced by one manufacturer, from the same materials and under essentially the same manufacturing conditions. Contained in any one lot number of canisters, no more than one lot number of chemical fill is permitted.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 Weight of fill. Sampling shall be conducted in accordance with MIL-STD-1235, CSP-1, code letter G.

4.4.3 Inspection procedure.

4.4.3.1 For examination. The sample canisters shall be examined in accordance with the classification of defects (4.4.3.3) and MIL-STD-105.

4.4.3.2 Weight of fill. The sample canisters shall be tested in accordance with 4.4.4.1 using an AQL of 1.0 percent defective.

4.4.3.3 Classification of defects.

(a) Canister, chemical projectile: 8 inch, QL, XM27 (Drawing D15-12-162) and canister, chemical projectile: 8 inch, TEP/DBA/EA (Drawing D15-12-213).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
	101	Weld plug position
	102	Weld plug height
	103	Burst disk damaged
	104	Workmanship (3.5)
Minor:	AQL 2.5 percent defective	
	201	Weight stamp missing or incorrect

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4.4.3.3 Classification of defects (continued).

(b) Canister, marking, chemical projectile: QL, XM27 (Drawing D15-12-184) and canister, marking, chemical projectile: TEP/DBA/EA (Drawing D15-12-214).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Marking missing or incorrect

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Weight of fill. The weight of fill shall be determined by weighing and marking the weights on unfilled canisters at the stated inspection interval (see 4.4.2.2). After filling and closing the sample canisters shall be weighed to the required accuracy to determine the net fill weight.

5. PACKAGING

5.1 Interplant shipment. Filled canisters shall be fitted with the same protective closure that was specified for the shipment of the empty metal canisters (caplug model B526 or equivalent). The protective closure shall be placed securely over the burst disk end of the canister to provide protection from foreign particles. Minimum additional packaging shall be used as required to assure that movement of the filled canisters from the filling area to the load, assembly and pack area is commensurate with economical material handling techniques and provides protection to the canisters from physical damage, corrosion and contamination.

6. NOTES

6.1 Intended use. This item is a component of the XM736 projectile.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) If simulant filled canisters are required (see 6.4).
(c) Preproduction.

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(1) Time allowed for contractor submission of samples for any Government tests and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Caution. Due to the hazardous nature of filling canisters with chemicals, contractors are advised to take all necessary precautions to protect personnel during filling.

6.4 Simulant. Simulant filled canisters shall be in accordance with Drawing D15-12-213.

6.5 Decontamination. Minor QL spillage can be cleaned from the exterior of the canister by using a sodium hypochlorite bleach solution and then flushing with clean water. Massive QL spillage will require cleaning with a solution of DS2 decontaminating agent (MIL-D-50030) and then flushing with clean water. Simulant spills may be cleaned from the exterior of the canister by large amounts of water.

6.6 QL filling by height. For production purposes, the canister filling operation may be process controlled by monitoring the fill height, provided the overall fill weight requirement is maintained. The Martron liquid level switch Model 14-03 is one type of liquid level monitor found suitable for checking height of liquid fill (Martron, Inc., 1626 East Maywood, Santa Ana, CA 92705). To perform the necessary calculations, the following QL density equation can be used:

Density QL (lbs/in^3) = .034346 - .0000178T
T = Temperatures (°F)

6.7 Simulant TEP/DBA/EA filling by height. For production purposes, the canister filling operation may be process controlled by monitoring the fill height, provided the overall fill weight requirement is maintained. The Martron liquid level switch Model 14-03 (see 6.6) is one type found suitable for checking height of liquid fill. To perform the necessary calculations, the following simulant density equation can be used:

Density simulant (lbs/in^3) = .034752 - .0000213T
T = Temperature (°F)

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6.9 Supersession data. This specification supersedes Chemical Systems Laboratory Purchase Description EA-C-1166, 14 April 1977.

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